

RATNAKSH TYAGI

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AI Engineer with 2+ years of experience building RAG architectures and state-aware conversational agents. Expert in prompt/context engineering and metadata-driven memory to optimize LLM reliability and token efficiency.

TECHNICAL SKILLS

- **Prompt & Context Engineering:** Prompt Engineering, Context Engineering, System Prompts, Tool Prompts, Memory Design, Multi-turn Conversational Logic, Token Optimization, Prompt Evaluation
- **LLMs & RAG Systems:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, FAISS, Vector Databases, Context Retrieval & Management, Provenance Grounding
- **Programming & AI Development:** Python, NLP, Flask, Streamlit, Groq API, Relevance AI
- **Data & Analytics:** SQL, MySQL, Pandas, NumPy, Data Analysis, Data Cleaning, KPI Monitoring
- **Platforms & Tools:** GCP, Git, VS Code

WORK EXPERIENCE

AI Engineer | Xapotech Systems Pvt Ltd, Noida

Nov,2023 - Present

- Designed and engineered system prompts, tool prompts, and layered context structures to improve output reliability, token efficiency, and response consistency in LLM-based applications.
- Built Retrieval-Augmented Generation (RAG) pipelines using LangChain and FAISS, enabling grounded, source-aware responses with clear provenance.
- Developed multi-turn, state-aware conversational agents with prompt chains and decision logic to handle complex user workflows.
- Implemented conversational memory and context management strategies to maintain continuity across sessions and interactions.
- Evaluated and refined prompts through iterative testing, edge-case analysis, and basic guardrails to reduce hallucinations and unsafe responses.

PROJECTS

Investor Prep AI | LLMs | Prompt Engineering | RAG | React

- Built an AI-powered pitch simulator using LLMs (Groq) to provide founders with real-time evaluation and investment readiness feedback.
- Engineered context-aware prompt logic to simulate realistic investor personas and generate tailored VC questions based on pitch content.
- Developed a multi-dimensional scoring engine that analyzes pitch quality and provides actionable metrics to increase funding probability.

RT Sales Copilot Agent | Prompt Engineering | Relevance AI

- Engineered system and task-specific prompts to analyze company websites and LinkedIn profiles for pre-sales intelligence.
- Designed prompt chains to extract, summarize, and contextualize business information into structured call preparation reports.
- Improved response relevance and consistency by refining context selection and prompt sequencing.

Smart Calendar Scheduling Agent | Python | Flask | NLP | LLM

- Built a conversational AI agent capable of managing calendar events through multi-turn natural language interactions
- Implemented intent recognition and context persistence to handle scheduling modifications, clarifications, and follow-up actions.
- Designed prompt logic for state-aware conversations and decision-making across user intents.

StudyPath: Study Abroad Platform | Next.js | Node.js | TypeScript | Prisma

- Developed an AI-assisted counselor leveraging Gemini APIs to provide personalized university recommendations based on complex academic profiles and geographic preferences.
- Engineered a real-time application tracking dashboard to manage critical milestones for 100+ global institutions, reducing process Friction through automated status updates.
- Built a robust full-stack architecture with a Next.js frontend and Node.js backend, optimizing data retrieval for large-scale university metadata and fee structures.

EDUCATION

Bachelor of Technology in Computer Science
Jaypee Institute of Information & Technology Noida-62

2019 – 2023

CERTIFICATIONS & ACHIEVEMENTS

- Generative AI - Udemy